

JINTAO Meng

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WORK EXPERIENCE

2020.9-Now **Associate Researcher**, Shenzhen Institute of Advanced Technology (SIAT), CAS, Shenzhen, China

Duty:

1. Teaching courses in Computer Science.
2. Guiding students to optimize scientific and DL applications on Mobile and PC devices.
3. Cooperating with Tencent AI infrastructure team for fast serving LLM models.

2017.2-2020.9 **Senior Research Engineer**, Tencent AI Lab, Shenzhen, China

Duty:

1. Leading a team working on accelerating DL models for moba games and AI healthcare.
2. Developing image recognition DL models for healthcare applications such as Pulmonary nodule and Colorectal cancer.

2008.7-2017.2 **Research Engineer**, Shenzhen Institute of Advanced Technology(SIAT), CAS, Shenzhen, China

Duty:

1. Pursuing PHD degree in CAS on Computer Architecture.
2. Cooperating with Argonne National Laboratory and BGI Genomics for developing bio-informatics applications such as Sequence Alignment and Assembler.

EDUCATION BACKGROUND

2016.7 Ph.D, COMPUTER ARCHITECTURE, INSTITUTE OF COMPUTING TECHNOLOGY, CAS, **Beijing**

Ph.D. Thesis: "Highly Scalable De Novo Genome Assembly"

Advisor: Xiaofang Zhao, Shengzhong Feng

2008.7 Master, COMPUTER SCIENCE, CENTRAL CHINA NORMAL UNIVERSITY, **Wuhan**

Master Thesis: "Energy Efficient Protocol Design for Wireless Sensor Networks"

Advisor: Liansheng Tan

2005.7 Bachelor, COMPUTER SCIENCE, CENTRAL CHINA NORMAL UNIVERSITY, **Wuhan**

SELECTED PUBLICATIONS

1. Yangjie Zhou, Honglin Zhu, Qian Qiu, Weihao Cui, Zihan Liu, Peng Chen, Mohamed Wahib, Cong Guo, Siyuan Feng, **Jintao Meng**, Haidong Lan Jingwen Leng, Yun Lin, Jinsong Dong, Wenxi Zhu,

Minwen Deng, Helix: Architecture-Agnostic Compilation Framework for High-performance Dynamic Tensors Computing, in Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2025), St. Louis, MO, USA, 2025

2. Cong Ma, Du Wu, Zhelang Deng, Jiang Chen, Xiaowen Huang, **Jintao Meng***, Wenxi Zhu, Bingqiang Wang, Amelie Chi Zhou, Peng Chen, Minwen Deng, Yanjie Wei, Shengzhong Feng, Yi Pan, NM-SpMM: Accelerating Matrix Multiplication Using N:M Sparsity with GPGPU, in Proceedings of the 39th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2025), June 3-7, 2025, Milan, Italy.
3. Xiaowen Huang, Renjie Mao, Lvfang Tao, Wenxi Zhu, and Minwen Deng, Xu Zhang, **Jintao Meng***, and Yanjie Wei, Nan Zhou, Amelie Chi Zhou, Bingqiang Wang, and Shengzhong, Parallel Code Generation with Large Language Model, IEEE ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP 2025) Workshop FCPC 2025, March 1-5, Las Vegas, Nevada, United States.
4. Du Wu*, **Jintao Meng***, Wenxi Zhu, Minwen Deng, Mohamed Wahib, Tao Luo, Yanjie Wei, auto-GEMM: Pushing the Limits of Irregular Matrix Multiplication on Arm Architectures, in Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2024), Atlanta, Georgia, Nov. 2024. (CCF A)
5. Zhen Ju, Jingjing Zhang, Xuelei Li, **Jintao Meng**, Yanjie Wei, SeedHit: a GPU friendly pre-align filtering algorithm, IEEE/ACM Transactions on Computational Biology and Bioinformatics(TCBB), 2024, 1(1): 1-9 (IF=4.5, JCR Q1)
6. Zhizhao Wang, Xuelei Li, Jianping Fan, **Jintao Meng**, Zhenli Lin, Yi Pan, Yanjie Wei, SWsnn: A Novel Simulator for Spiking Neural Networks, **Journal of Computational Biology**, 2023, 30(9): 951-960 (IF=1.7, JCR Q2)
7. Huaipeng Zhang, Nhut-Minh Ho, Yigit Polat Dogukan, Peng Chen, Mohamed Wahib, Truong Thao Nguyen, **Jintao Meng**, Rick Siow Mong Goh, Satoshi Matsuoka, Tao Luo, Weng-Fai Wong, Simeuro: A Hybrid CPU-GPU Parallel Simulator for Neuromorphic Computing Chips, **IEEE Transactions on Parallel and Distributed Systems**, 2023 (IF=4.18, JCR Q1)
8. Tao Shen, Fuxu Liu, Zechen Wang, Jinyuan Sun, Yifan Bu, **Jintao Meng**, Weihua Chen, Yuguang Mu, Weifeng Li, Guoping Zhao, Sheng Wang, Yanjie Wei, Liangzhen Zheng, zPoseScore model for accurate and robust protein-ligand docking pose scoring in CASP15, **PROTEINS: Structure, Function, and Bioinformatics** (IF=4.088, JCR Q2, CASP 15 invited paper, rank 2 on protein-ligand complexes prediction)
9. Lingqi Zhang, Mohamed Wahib, Peng Chen, **Jintao Meng**, Xiao Wang, Toshio Endo, Satoshi Matsuoka, PERKS: a Locality-Optimized Execution Model for Iterative Memory-bound GPU Applications, 37th ACM International Conference on Supercomputing (ICS 2023), June 2023, Orlando, USA (CCF B)
10. Lingqi Zhang, Mohamed Wahib, Peng Chen, **Jintao Meng**, Xiao Wang, Toshio Endo, Satoshi Matsuoka, Revisiting Temporal Blocking Stencil Optimizations, 37th ACM International Conference on Supercomputing (ICS 2023), June 2023, Orlando, USA (CCF B)
11. Lingqi Zhang, Mohamed Wahib, Peng Chen, **Jintao Meng**, Xiao Wang, Toshio Endo, Satoshi Matsuoka, Exploiting Scratchpad Memory for Deep Temporal Blocking, The 15th Workshop on General Purpose Processing Using GPU (GPGPU 2023), Feb 2023, MONTREAL, CANADA.

12. Liangzhen Zheng, **Jintao Meng***, Mingzhi Lin, Rui Lv, Hongxi Cheng, Lixin Zou, Jinyuan Sun, Linxian Li, Ruobing Ren, Sheng Wang, Structure Prediction of the whole Proteome of Monkeypox variants, in **Acta Materia Media**, 2022.
13. Haidong Lan, Wenxi Zhu, Qian Qiu, Dou Wu, Honglin Zhu, Jingjing Zhao, Xinghui Fu, Minwen Deng, **Jintao Meng***, Efficient Phase-Functioned Real-time Character Control in Mobile Games: A TVM Enabled Approach, in Proceeding of the 45th International Conference on Parallel Processing (**ICPP 2022**), Bordeaux, France. (Accept rate: 19%, CCF B)
14. Ju Zhen, Huiling Zhang, **Jintao Meng**, etc, nGIA: A novel Greedy Incremental Alignment based algorithm for gene sequence clustering, **Future Generation Computer Systems**, 2022 (IF 7.187, JCR Q1)
15. Liangzhen Zheng, **Jintao Meng**, etc, Improving protein–ligand docking and screening accuracies by incorporating a scoring function correction term, **Briefings in Bioinformatics**, 2022, 23(3) (IF 11.6, JCR Q1)
16. **Jintao Meng**, Peng Chen, Mohamed Wahib, Mingjun Yang, Liangzhen Zheng, Yanjie Wei, Shengzhong Feng, Wei Liu, Boosting the predictive performance with aqueous solubility dataset curation, **Scientific Data**, 2022, 9(1): 1-13 (IF 6.101, JCR Q1)
17. **Jintao Meng**, Chen Zhuang, Peng Chen, Mohamed Wahib, Bertil Schmidt, Xiao Wang, Haidong Lan, Dou Wu, Minwen Deng, Yanjie Wei, Shengzhong Feng, Automatic Generation of High-Performance Convolution Kernels on ARM CPUs for Deep Learning, **IEEE Transactions on Parallel and Distributed Systems**, 2022, 33(11): 2885-2899 (IF 4.18, JCR Q1)
18. Huiling Zhang, Ying Huang, Zhendong Bei, Zhen Ju, **Jintao Meng**, etc, Inter-Residue Distance Prediction From Duet Deep Learning Models, **Frontiers in Genetics**, 2022, 13 (IF 4.274, JCR Q2)
19. Jianqiu Ge, **Jintao Meng***, Ning Guo, Yanjie Wei, Pavan Balaji, Shengzhong Feng, Counting kmers for biological sequences at large scale, **Interdisciplinary Sciences: Computational Life Sciences**, 2020, 12(1): 99-108 (IF 4.274, JCR Q2)
20. Haidong Lan, **Jintao Meng***, Christian Hundt*, Bertil Schmidt, Minwen Deng, Xiaoning Wang, Weiguo Liu, Yu Qiao, Shengzhong Feng, FeatherCNN: Fast Inference Computation with TensorGEMM on ARM Architectures, **IEEE Transactions on Parallel and Distributed Systems**, pp. 580-594, March, 2020. (IF 4.18, JCR Q1)
21. **Jintao Meng**, Sangmin Seo, Pavan Balaji, Yanjie Wei*, Bingqiang Wang, Shengzhong Feng, SWAP-Assembler 2: Optimization of De Novo Genome Assembler at Extreme Scale, in Proceeding of the 45th International Conference on Parallel Processing (**ICPP 2016**), Philadelphia, PA. (Accept rate: 18%, CCF B)
22. **Jintao Meng**, Bingqiang Wang, Yanjie Wei*, Shengzhong Feng*, Pavan Balaji. SWAP-Assembler: Scalable and Efficient Genome Assembly towards Thousands of Cores, **BMC Bioinformatics**, Vol. 15 Supplement 9. (IF 3.0, JCR Q1)
23. **Jintao Meng**, Bingqiang Wang, Yanjie Wei*, Shengzhong Feng*, Pavan Balaji. SWAP-Assembler: Scalable and Efficient Genome Assembly towards Thousands of Cores, in 4th annual RECOMB satellite workshop on massively parallel sequencing (RECOMB-seq 2014), April, 2014. (CCF B)

ACADEMIC ACTIVITIES

Reviewer of Journals and Conferences:

Transaction on Parallel and Distributed Systems (**TPDS**)

Journal of Parallel and Distributed Computing (**JPDC**)

Transactions on Knowledge and Data Engineering (TKDE)
 Transactions on Systems, Man, and Cybernetics (TSMAC)
 Journal of Computer Science and Technology (JCST)
 BMC Bioinformatics, The Journal of Supercomputing
 ICPP 2022 CCGrid 2020, SC 2019 , PPOPP 2019 , IPDPS 2018

Academic Talks:

- (a) *Poster*: FeatherCNN: Fast Inference Computation withTensorGEMM on ARM Architectures, in **Supercomputing 2018**, Dallas, TX.
- (b) *Poster*: SWAP-Assembler 2: Scalable Genome Assembler towards Millions of Cores - Practice and Experience, in ATIP workshop in **Supercomputing 2015**, Austin, TX.
- (c) *Oral Talk*: FeatherCNN: Fast Inference Computation withTensorGEMM on ARM Architectures, in Alsys workshop in **SOSP 2017**, Shanghai, China.
- (d) *Oral Talk*: SWAP-Assembler: Scalable and Efficient Genome Assembly towards Thousands of Cores, in **CNCC 2018**, Hangzhou, China.

SELECTED FUNDINGS

- (a) National Science Fund China (NSFC) for Young Scholars , 2009A091100017, Highly Scale-able Genome Assembly to Millions of Cores, 20017/12-2020/12, 240,000 RMB, PI
- (b) National High-tech RD Program of China (863 Program), 2014AA01A302, High Performance Software Programming Framework and Applications for Exa-scale - sub-project (Research on Highly Scale-able Graph Algorithms) , 2015/06-2018/06
- (c) Fundamental Research in Shenzhen, Research on Efficient Large Scale Meta-Genome analysis, 2014/12-2017/12, 200,000 RMB
- (d) National Development and Reform Commission (NDRC), Health Service Platform based on Peta-Scale Genome Data Processing, 2014.2-2017.2, 1.6 million RMB
- (e) Guangdong Province-CAS Comprehensive Strategic Cooperation, 2009A091100017, High Throughput Genome Sequencing Technology and its Applications, 2009/02-2011/06, 1.2 million RMB

HONORS AND CERTIFICATES

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| 2025 | 1st Place, Fastcode Programming Challenge 2025, hold by IEEE/ACM SIGPLAN PPOPP 2025 |
| 2021, 2023, 2025 | Outstanding staff of Shenzhen Institutes of Advanced Technology, CAS |
| 2022 | 2st Place in predicting protein-ligand complexes in CASP15 |
| 2021 | National Champion in the ACM-IPCC International Parallel Application Challenge |
| 2018 | Tencent Five-Star Employee |
| 2010,2012,2013,2015 | Outstanding staff of Shenzhen Institutes of Advanced Technology, CAS |
| 2005, 2008 | Outstanding graduates in Central China Normal University |
| 2003, 2004 | ACM/ICPC-Asia, rank 36, 27, 10 in Guangzhou, Shanghai, and Beijing site |
| 2002, 2003, 2004 | Outstanding Students in Central China Normal University |